



Safe Digging Essentials

A free LSBUD Insight guide for anyone commissioning, planning, managing or carrying out work that may disturb the ground.

Core message

Plan the work, locate and identify services, then excavate safely.



Who this guide is for: householders, contractors, surveyors, designers, managers, councils, estates teams and anyone who needs a simple starting point before ground is broken.

Important: this guide is not a substitute for legal duties, competent advice, asset owner requirements, site-specific risk assessment or HSE guidance.

1. The essential safe digging principle

Every excavation, however small, should begin with the assumption that buried services may be present. Plans, records, site observations, detection equipment and trial holes all have a role. None should be treated as a single, complete answer on their own.

The Health and Safety Executive's HSG47 guidance is aimed at people who commission, plan, manage or carry out work on or near underground services, and it identifies three basic elements of a safe system of work: planning the work, locating and identifying buried services, and safe excavation.



What “safe digging” means in practice

- **Before the site:** understand the task, site history, asset owners, work limits, emergency arrangements and whether the activity involves ground disturbance.
- **Before the excavation:** obtain and review utility information, mark known and suspected routes, and decide whether detection, survey or trial holes are needed.
- **During the excavation:** expose services carefully, control mechanical excavation, protect identified services and stop if conditions differ from expectations.
- **After the work:** record what was found, update drawings where appropriate and capture lessons for future works.

LSBUD Insight recommendation

If the work may disturb the ground, consider starting with OneCall Utility Search, then use Top-Up, survey or report options where the site risk justifies.

2. The safe digging workflow

Use this workflow as a simple control structure. It is deliberately written in plain English so it can be used by householders, managers and site teams.

Stage	What to do	Why it matters
Plan the work	Define the task, location, depth, method, programme, responsibilities and emergency arrangements. Check whether the work is planning, maintenance, construction, highway, utility or environmental in nature.	Risk is easier and cheaper to control before a team is on site and under programme pressure.
Obtain information	Gather relevant utility search information and any known site records. Consider private services, previous works, estate records, highways records and asset owner requirements.	Plans and records create the first picture of what may be present, but they do not prove exact position or depth.
Locate and mark	Use competent locating methods and mark routes on the ground where appropriate. Consider PAS128/GPR survey where uncertainty or consequence is high.	A plan that has not been interpreted on site can lead to false confidence.
Excavate safely	Use safe digging methods, expose services carefully, control plant near services and stop if the site differs from the expected picture.	Most serious incidents happen when assumptions replace verification.
Review and record	Record exposed services, unexpected findings, near misses and changes that would help the next person planning work at the site.	Good records reduce repeat risk and improve future searches, packs and briefings.

A practical rule: **if the work description changes, the risk assessment and search/survey decision may need to change too.**

3. Key risks to consider

The examples below are not exhaustive, but they help users recognise why safe digging is relevant to very different types of work.

Risk area	Typical warning signs	Good practice response
Electricity cables	Substations, service pillars, lighting columns, cabinets, ducts, previous repairs or cable markers.	Treat every cable as live unless competent evidence says otherwise. Use plans, locating equipment, careful exposure and asset owner instructions.
Gas pipes	Gas meters, valve covers, pipeline markers, service routes into buildings, known utility corridors.	Avoid mechanical excavation close to suspected routes until controls are agreed. Stop work and follow emergency procedures if damage or smell is suspected.
Water pipes and sewers	Stop taps, chambers, gullies, drainage runs, wet ground, manholes, older property connections.	Expect service connections and private drainage where records are weak. Use careful exposure and consider CCTV/drainage survey where relevant.
Telecoms and fibre	Cabinets, chambers, ducts, footway routes, poles, service entries to premises.	Telecoms damage can be operationally significant even where direct injury risk is lower. Protect routes and use appropriate location methods.
Private or unrecorded services	Outbuildings, external lighting, pumps, gates, signage, historic installations, private estates.	Do not assume all assets appear in standard utility records. Ask site owners and use local knowledge, detection and trial holes.

Stop-work trigger

If the excavation exposes an unexpected service, damaged service, unusual material, warning tape or unexplained duct, stop and reassess before

4. Choosing the right next step

Safe digging decisions should be proportionate to the work, the uncertainty and the consequences of error. A householder installing a short fence and a contractor planning deep linear works do not need the same journey, but both need the right prompt at the right time.

If the user needs...	Likely route	Access mode
Plain-English advice	Read Safe Digging Essentials, Do I Need a Utility Search?, Service Connection Risk Guide or relevant FAQ/Wiki entries.	Guest
Utility asset information	Start OneCall Utility Search. Use Top-Up where additional asset owner coverage or evidence is required.	Login for submission
Higher confidence in position	Request Select Surveys / PAS128 / GPR quote where plans are unclear, consequence is high or design/excavation depends on utility position.	Login for quote
Large or complex site	Use Large Site Search or Comprehensive Premium where standard search size, complexity or evidence requirements are exceeded.	Login / Pro
Watercourse or environmental site risk	Use Open Water Proximity Risk Report or Environmental Constraints Report alongside relevant guidance.	Login for report
Team or repeat-work governance	Use Pro Account, training, toolbox talks, history, saved projects and repeat search workflow.	Pro

This is the logic behind the LSBUD Insight Library: free content helps the user understand the risk; the recommendation engine then routes them into OneCall, survey, report, quote or Pro account options only where appropriate.

5. Quick checklist before the ground is broken

Check	Question	Done
Scope	Is the exact work area, depth, method and programme understood?	■
Search	Has a utility search been considered or completed for the work area?	■
Private services	Could private, estate, historic or service connection assets be present?	■
Site markings	Have known and suspected routes been identified, marked and briefed?	■
Survey need	Is PAS128/GPR, CCTV drainage or another survey needed to reduce uncertainty?	■
Excavation method	Are safe digging methods, plant exclusion zones and hand-dig controls agreed?	■
Emergency plan	Does the team know what to do if a service is exposed, struck, damaged or suspected?	■
Record	Will unexpected findings, exposed services and lessons be recorded?	■

Simple decision prompt

If you are unsure whether the information is good enough to dig safely, pause and escalate. A search, survey or competent review is cheaper than

6. Glossary

Term	Meaning
Asset owner	Organisation that owns or operates underground apparatus such as cables, pipes, sewers, ducts or pipelines.
OneCall Utility Search	The core LSBUD utility search route for obtaining relevant utility asset information for a defined area.
Top-Up Search	Additional search route where broader or supplementary coverage is required beyond the core search context.
PAS128	A British Standards specification for underground utility detection, verification and location. In practice it helps clients specify survey levels and outputs.
GPR	Ground Penetrating Radar, one detection technique that may be used as part of a utility survey.
Trial hole	A controlled excavation used to confirm the position and depth of known or suspected services.
WKT	Well-Known Text, a text format for describing site geometry such as polygons and lines.

Next Library PDFs recommended after this guide: **Do I Need a Utility Search?**, **Service Connection Risk Guide**, **Fencing and Gates Guidance**, **HSG47 Plain-English Summary** and **PAS128 Summary Guide**.

Sources and limitations

This guide is original LSBUD Insight library content. It is designed to summarise safe digging principles in plain English and to guide users toward suitable LSBUD Insight recommendations. It does not reproduce HSG47 or any third-party guidance in full.

Source	How it informed this guide
Health and Safety Executive, HSG47: Avoiding danger from underground services, Third edition, 2014.	Used as the principal reference for the three-part safe system: planning the work, locating and identifying buried services, and safe excavation.
Health and Safety Executive, Excavation and underground services.	Used to support the emphasis on managing risks, obtaining plans or suitable information, planning work and taking precautions around underground electrical cables.
LSBUD Insight Persona Engine and Library Catalogue.	Used to align the guide with future recommendations for OneCall Utility Search, Top-Up, Select Surveys/PAS128, Open Water, Environmental Constraints, history and Pro account routes.

Limitations: this document is general guidance only. Users must consider legal duties, client requirements, CDM duties where relevant, current HSE guidance, asset owner instructions, site-specific hazards, competent advice and local operating procedures.

Recommended action: if works may disturb the ground, use the LSBUD Insight Hub to access relevant library content and consider the OneCall Utility Search route or other recommendation pathways suited to the site and activity.